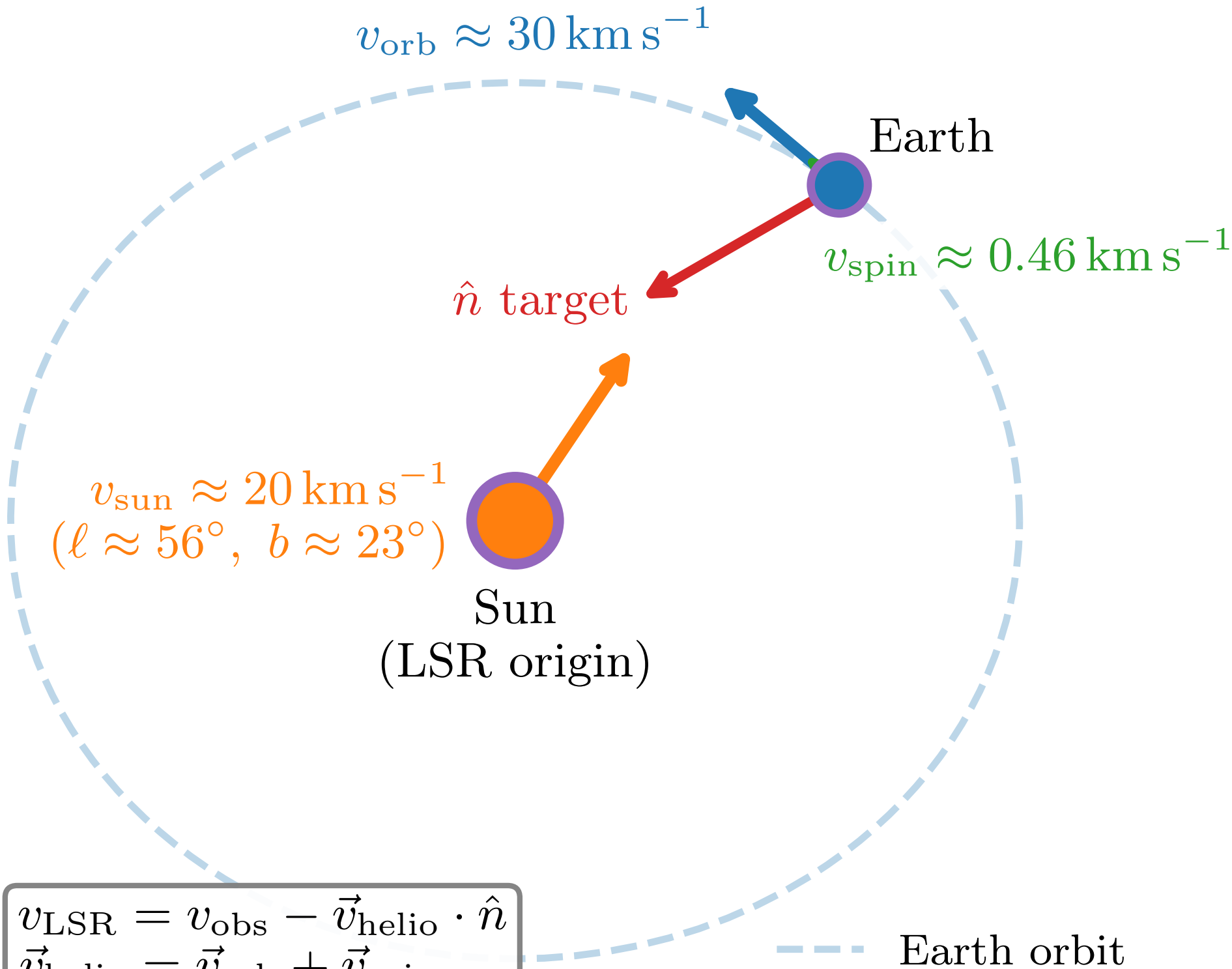

$$v_{\rm LSR} = v_{\rm obs} - \vec{v}_{\rm helio} \cdot \hat{n}$$
$$\vec{v}_{\rm helio} = \vec{v}_{\rm orb} + \vec{v}_{\rm spin}$$